

Question 9

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Which of the following statements about message-passing networks (MPNNs) are true?

- ☐ a. Both node-level and graph-level prediction tasks can be approached with MPNNs.
- ☐ b. For graph-level prediction tasks, the read-out function of MPNNs should be invariant to permutations of the nodes.
- ☐ c. MPNNs solve the graph isomorphism problem in linear time.
- ☐ d. The main calculation steps of MPNNs contain A) a message-generating phase and B) a vertex update phase.

[Check](#)**Question 8**

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Which of the following statements about generative models for small molecules are true?

- ☐ a. The syntax of SMILES strings cannot be learned from data. Long short-term memory networks trained on SMILES representations will produce <5% valid SMILES strings.
- ☐ b. Autoregressive models can be used to generate SMILES representations of small molecules.
- ☐ c. It is impossible to use Deep Learning method to generate molecular graphs.
- ☐ d. Variational autoencoders cannot be used to generate small molecules.

[Check](#)**Question 7**

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Which of the following statements about the U-Net architecture are correct?

- ☐ a. To aid the decoding process, feature maps from the encoder are concatenated with feature maps of the corresponding up-sampling level of the decoder.
- ☐ b. The U-Net was the first vision model where the encoder is based on self-attention instead of convolutions.
- ☐ c. The U-Net has been specifically developed for the object detection task.
- ☐ d. The U-Net architecture uses a symmetrical encoder and decoder.

[Check](#)**Question 6**

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Which of the following statements about the AlphaFold2 system are correct?

- ☐ a. AlphaFold2 is trained using deep reinforcement learning where the reward is given by the distance between the predicted and the ground-truth protein structure.
- ☐ b. AlphaFold2's structure module predicts both translation and rotation for each residue.
- ☐ c. The Global Distance Test (GDT) is a distance metric to measure the similarity between two given 1D amino-acid sequences.
- ☐ d. The Frame Aligned Point Error (FAPE) ensures that the loss between the predicted and the ground-truth structure does not change when the proteins are rotated.

[Check](#)

Question 5

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Which of the following statements about machine learning for chemical reactions and synthesis are true?

- ☐ a. Machine learning methods that predict outcomes of chemical reactions can be trained on databases containing known chemical reactions.
- ☐ b. There are methods that rely on so-called reaction templates, which are transformation rules for molecules.
- ☐ c. Transformer architectures have been used to predict outcomes of chemical reactions.
- ☐ d. Chemical reactions are determined by chaotic rules that cannot be learned with machine learning approaches.

Question 4

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Which of the following statements about evaluation of QSAR and drug target prediction methods are true?

- ☐ a. For evaluating prediction tasks for small molecules, neither a validation nor a test set is necessary, because the prediction quality on the training set is sufficiently informative.
- ☐ b. Because many of the classification problems are highly imbalanced, the accuracy is not an appropriate metric to compare QSAR and drug target prediction methods.
- ☐ c. Training datasets of small molecules often contain a *compound series bias*, that is sets of molecules with the same chemical backbone that just differ in the attached functional groups.
- ☐ d. Standard metrics used in machine learning (accuracy, AUC, mean-squared error, sensitivity & specificity, precision & recall) cannot be used because those metrics cannot use molecules as input.

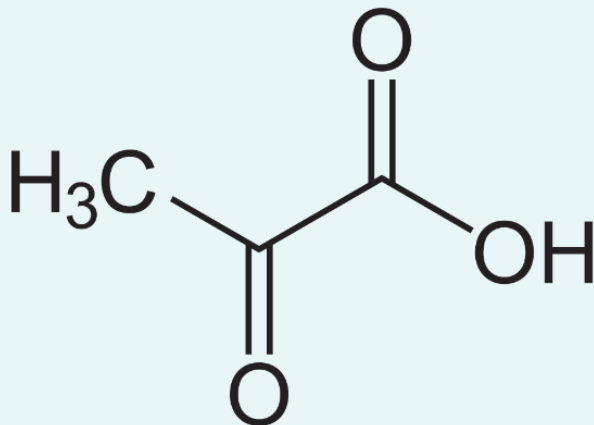
Question 3

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You are given the following molecule:



Which of the following SMILES strings encodes this molecule?

- ☐ a. O=C(C(=O)O)C
- ☐ b. CCCCOO
- ☐ c. CC(=O)C(=O)O
- ☐ d. OCCOO

Question 2

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Which of the following statements about drug target prediction and virtual screening are true?

- a. Virtual screening is a procedure in which a library of molecules is searched by a computational method.
- b. Machine learning methods can be used for virtual screening.
- c. A drug target prediction method predicts whether a small molecule can interact with a drug target.
- d. Tanimoto similarity to a known active molecule can be used for virtual screening

[Check](#)**Question 1**

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Assume you have a wet-lab procedure that a) involves a cell line and b) by which you can determine how a small molecule can inhibit the function of the protein EGFR. Your aim is to find small molecules that inhibit the function of this protein as best as possible. You can assume that substantial amounts of data on inhibitors of this protein is contained in public databases. Which of the following statements are true?

- a. Someone suggests to train a machine learning model that discriminates between EGFR inhibitors and non-inhibitors on public data and then use this model to select molecules with high prediction scores for wet-lab testing. This strategy is **highly unlikely** to increase the average pIC50 values of the tested molecules.
- b. You want to select hundred molecules for testing with this bioassay. Someone suggests to check the ChEMBL database for already known EGFR inhibitors instead of testing randomly selected molecules. This strategy is **likely** to increase the average pIC50 values of the tested molecules.
- c. You have measured two molecules with this bioassay, the first with a pIC50 value of 6.2 and another with a pIC50 value of 8.1. This means that the first molecule has a stronger inhibitory effect on EGFR.
- d. The procedure can be considered as an **in-vitro bioassay**.

[Check](#)**Question 10**

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Which of the following statements about Multiple Instance Learning (MIL) are true?

- a. MIL-based machine learning methods should be able to process bags containing variable numbers of elements.
- b. Instance-level scoring first aggregates a model's representations via pooling and then applies the output layer to the pooled representation.
- c. MIL-based machine learning methods should always produce different outputs for the same bag when the order of the elements within the bag is shuffled.
- d. The witness rate refers to the fraction of samples indicating the right class.

[Check](#)